MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

Artificial Intelligence and Society

Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 4.3: Generative Language Models

Jana Lasser
Research group
Complex Social & Computational Systems (CS)²

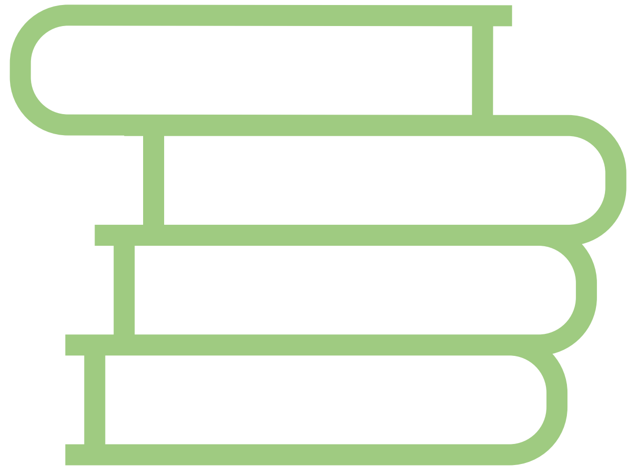
IDea_Lab
Das Interdisziplinäre Digitale
Labor der Universität Graz

UNI GRAZ

Generative Language Models

- Generative language models – sometimes also "Large Language Models" or "LLMs" – can produce coherent sounding text on demand.
- They have become extremely widespread since the release of the chat bot ChatGPT in 2022.
- To train generative language models, all types of machine learning are needed.

Step 1: Pre-training with Self-Supervised Learning



giant text corpus

GPT-3's training corpus includes:

- CommonCrawl (400 billion tokens, 46% English) – low quality.
- Reddit (19 billion tokens), Wikipedia (3 billion tokens) – high quality.
- Books (67 billion tokens) – high quality.

legal aspects of AI

See also Brown et al., Language Models are Few-Shot Learners, *arXiv* (2020).

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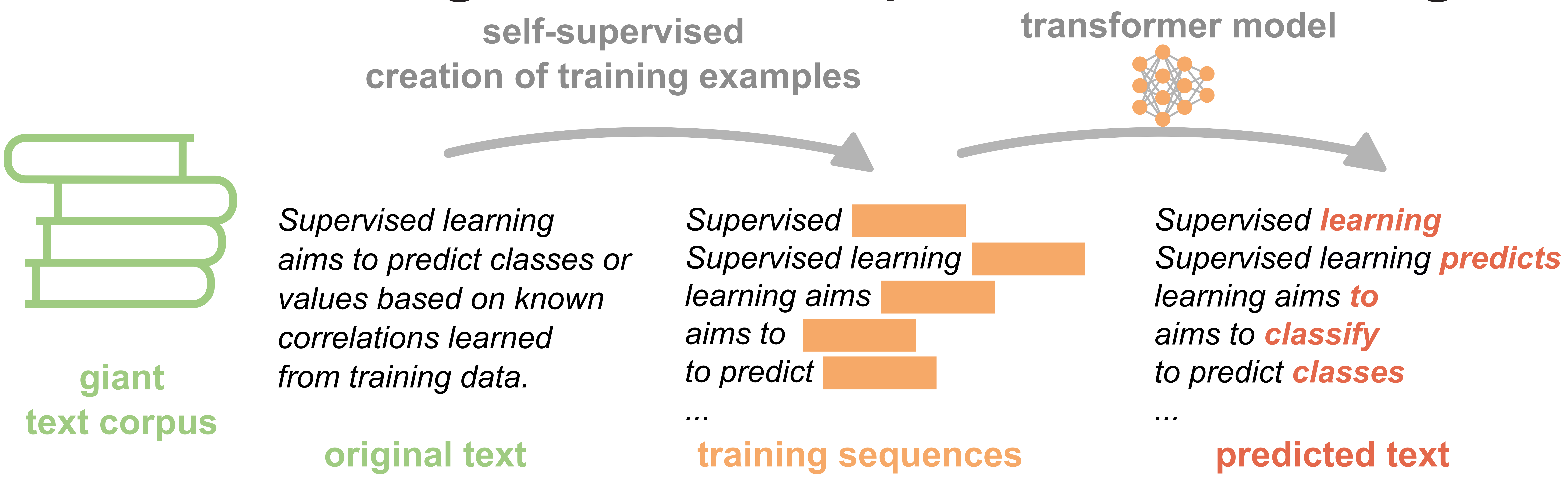

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- CommonCrawl (400 billion tokens, 46% English) – low quality.
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Cost of training GPT-4:
$100 mio

economic aspects of AI

See also Brown et al., Language Models are Few-Shot Learners, *arXiv* (2020).

Pre-trained Language Model

- A language model pre-trained in this way is already very good at generating **coherent sequences of words.**
- It "speaks" the languages sufficiently present in the training dataset and has learned syntax and grammar structures.
- Pre-trained language models are **very large**: they are neural networks with up to 671 billion parameters* (weights). This corresponds to about 1.5 TB of hard drive space just for storing the model weights.
- However, the model cannot yet **follow instructions**. Produced text follows the structure of books or articles rather than conversations.

*Size of the most recent DeepSeek model V3.

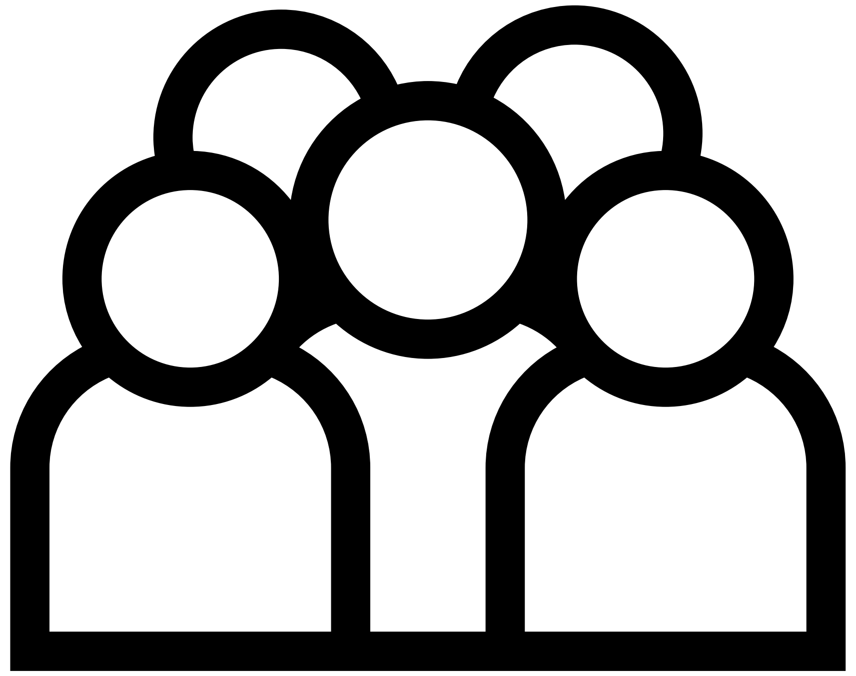
Step 2: Supervised Learning for Instruction Following

list with instructions (prompts)

Explain step by step ...

→

humans produce text as output



→

training data

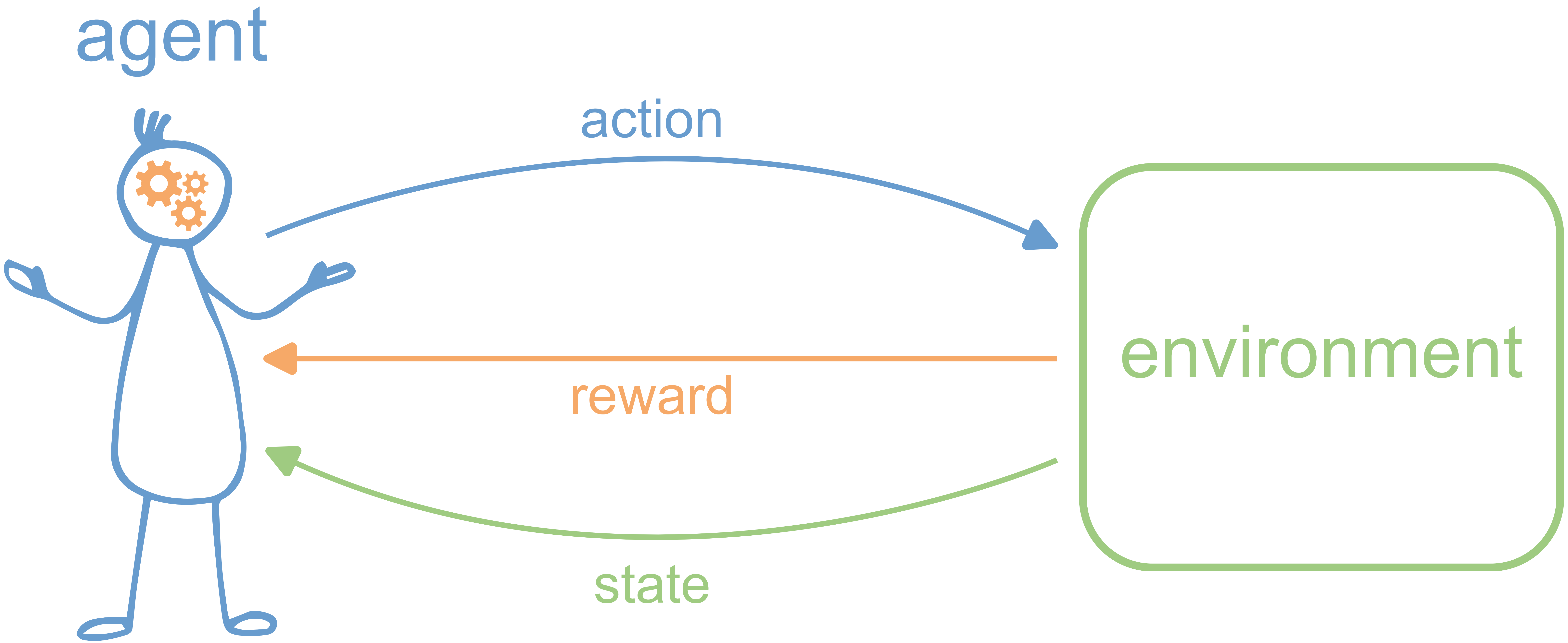
prompt	human outputs
explain ...	First, ...
tell me ...	Once upon a time...
enumerate ...	1. ... 2. ...

preceding conversation + prompt → generated output sequence

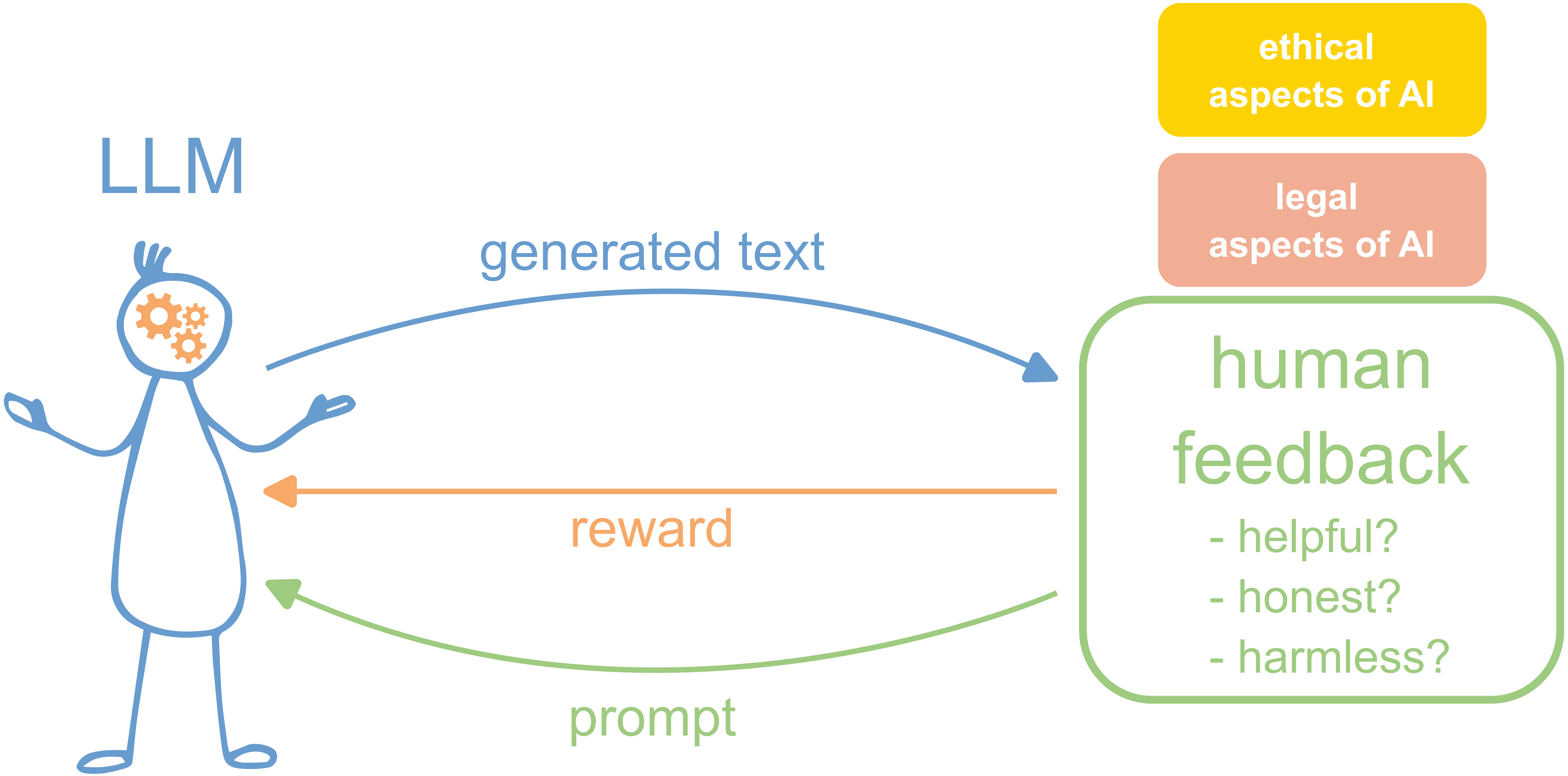
Instruction-Tuned Language Model

- The "fine-tuning" of the model with a small dataset of instruction-response pairs changes the model's weights only superficially – the basic structure of the language remains intact.
- An instruction-tuned language model can respond meaningfully to instructions as an input sequence.
- However, it can still produce answers that may be **unfriendly, dangerous, or misleading**.

Review: Reinforcement Learning

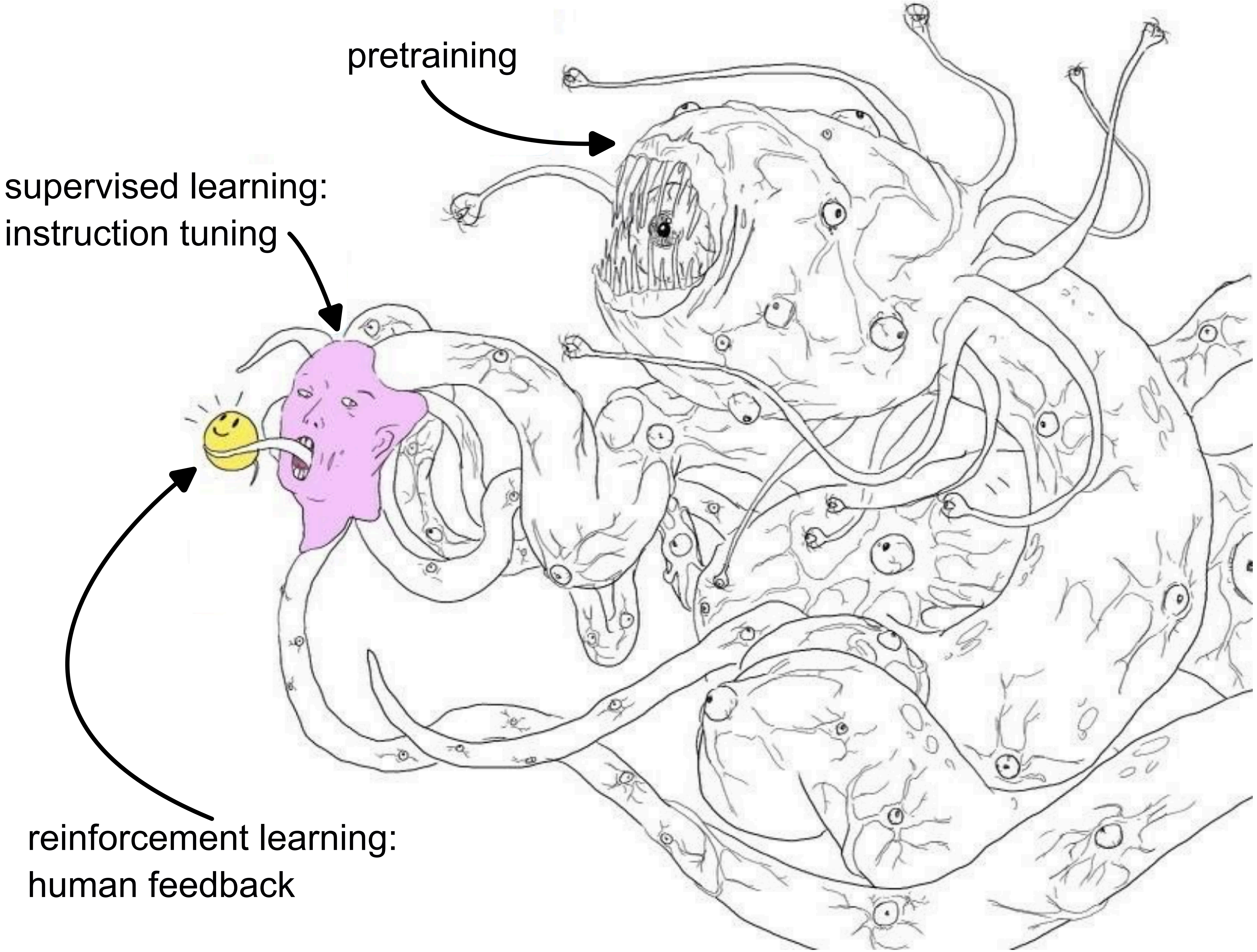


Review: Reinforcement Learning



Step 3: Reinforcement Learning with Human Feedback

prompt		**answer**	**feedback**
How is the weather outside?	→ LLM →	*The weather outside is very nice!*	not helpful
Can coughing cause a heart attack?	→ LLM →	*Yes, coughing can cause a heart attack.*	not honest
Make me a plan to rob a bank.	→ LLM →	*Here is a detailed plan on how to rob a bank ...*	not harmless

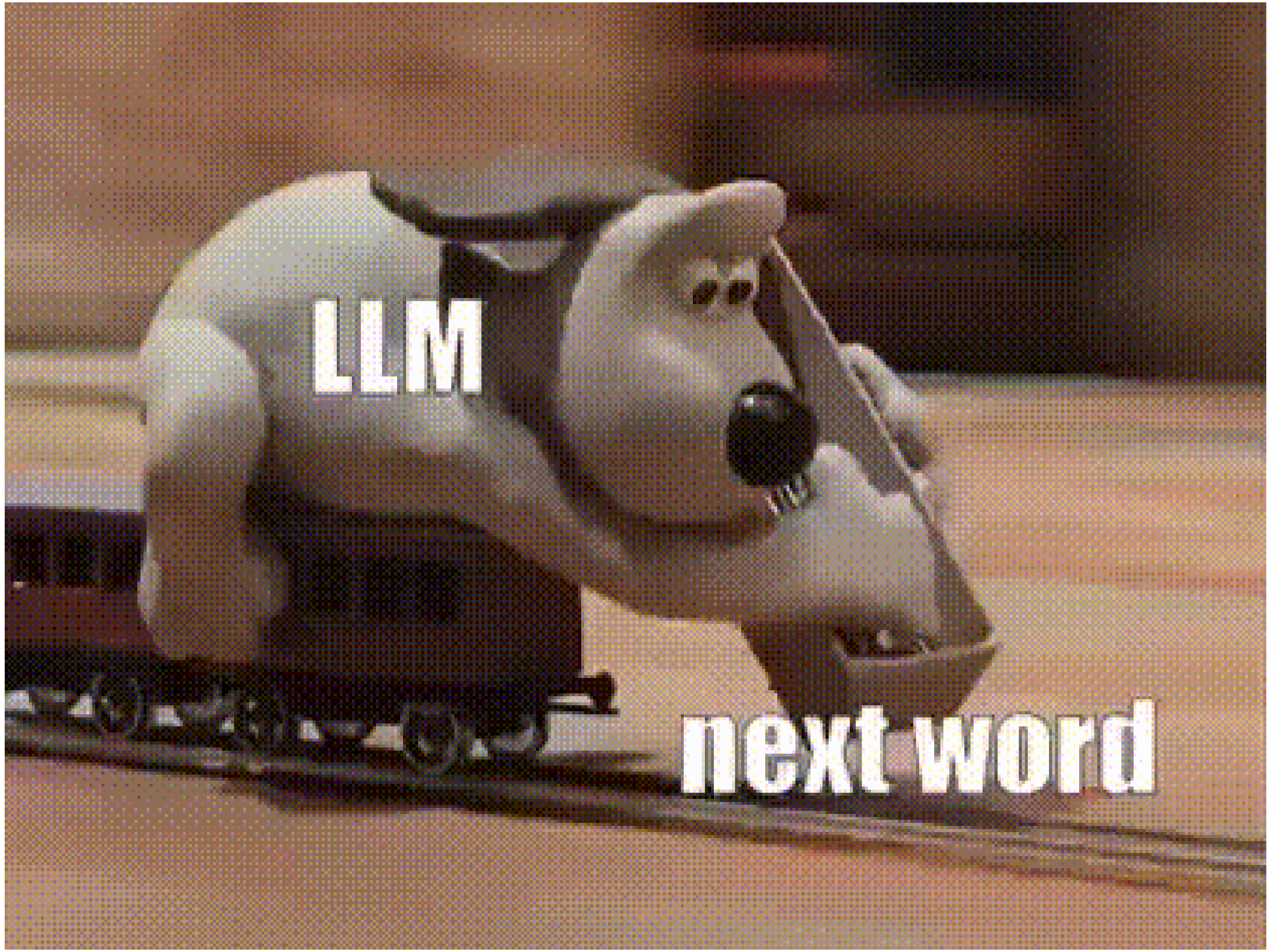


- Reinforcement learning and fine-tuning only superficially change the model's weights.
- Beneath it still lies the pre-trained model.
- Text patterns learned in pre-training can still **come to light** with the right prompting – even if they are undesirable.

Figure adapted from: Samim, alignment memes.

"Halluzinations"

What an LLM like ChatGPT does under the hood.



Where do LLMs work (not) well?

- Tasks that tolerate errors: Generation of stories, editing of existing text, ...

  → **LLMs are very good at producing coherent-sounding texts.**

- Tasks requiring factual knowledge: Knowledge retrieval, summarizing, ...

  → **LLMs have no concept of factuality.**

- Tasks requiring critical thinking: Brainstorming, philosophical debates, ...

  → **LLMs are trained to be helpful and harmless. They are reluctant to contradict.**

- Tasks requiring logic: Mathematics, coherent reasoning, creating a plan, ...

  → **LLMs have no understanding of reality.**